Test Specification (참조용으로 작성된 자료)

Test Sample 1. Undervoltage and Overvoltage Test

Purpose When an overvoltage or undervoltage has been detected, the DUT enters a safe mode, i.e. no undefined function shall occur during an overvoltage or undervoltage phase.

When the DUT returns into the operating voltage range again, it automatically fulfills all functions as specified again.

Functional degradations shall be defined in the drawing or in the specifications.

Test

Operating mode	Operating mode II.c
Test duration	Voltage increments at an interval of t_test = 10s
Test voltage	The voltage V_min shall be reduced incrementally by a value of ΔV_test = 0.5V until the undervoltage shutoff circuit is triggered and the motor is switched off. Then the voltage shall be increased incrementally again by a value of ΔV_test = 0.5V until the motor is switched on again. The voltage V_max shall be increased incrementally by a value of ΔV_test = 0.5V until the overvoltage shutoff circuit is triggered and the motor is switched off. Then the voltage shall be reduced incrementally again by a value of ΔU_test = 0.5V until the motor is switched on again.
No. of cycles	1

A suitable hysteresis of ≤ 0.5V shall be used at the transition from the undervoltage range or overvoltage range to the operating voltage range.

Requirement

After the test the DUTs shall automatically return to operating mode II.c

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분류 : Electrical Test

간단 설명 : BLDC Motor 정지 전압 확인 평가

동작 기준 저전압 기준으로 동작 _0.5V 낮추며 정지되는 점을 찾는다.

동작 기준 고전압 기준으로 동작 _0.5V 올리며 정지되는 점을 찾는다.

Test Specification

Test Sample 2. On & Off Test

Purpose

The safe initialization, start-up, and shutdown of the component shall be demonstrated. A sufficient cycle reserve of memories, capacitors, transformers, and relays shall also be demonstrated.

Test procedure

Cycle: One cycle is 1min operating mode II.a / and 1min operating mode II.c

No. of cycles:

18,000 cycles with the following distribution:

- 10,000 cycles at TRT / 4,000 cycles at Tmax / 4,000 cycles at Tmin

Test duration:

Approx. 600 h. In one cycle, all connected supply inputs switched in the vehicle shall be switched on and off

Requirement

The state defined for operating mode II.a is reached in each cycle. The data/parameter integrity must be checked every 18,000 cycles as a minimum.

분류 : Electrical Test / Endurance Test

간단 설명 : ON/OFF 동작을 반복하며 전기적 성능 유지, 구조적 성능 유지를 확인한다.

1 분 간격 ON/Off 동작을 수행한다 : 상온에서 10,000 회 / 고온에서 4,000 회 / 저온에서 4,000 회

Test Sample 3. Water Test

Test

The test shall be performed in accordance with ISO 20653 with the following parameters:

Operating mode	Operating mode II.c
Test duration	According to ISO 20653.
Test voltage	12V
No. of cycles	According to ISO 20653.

Requirement

The degree of protection as per ISO 20653 stipulated in the component specifications shall be reached.

분류 : 부식성 평가

간단 설명 :

ISO 20653 규격 _ 차량 내 장착 위치에 따라 IP CODE 가 분류되며,

Hyoseong Blower Motor 는 일반적으로 IP2K / IP4K Water test 진행.

IP CODE 에 따라 시험 조건이 다르며 ISO 20653 규격은 Water 시험법을 설명하는 데에 집중하여 작성되어 있음.

Test Specification

Test Sample 3. Random Vibration Test

Test

Test shall be conducted according to the Random Vibration test described in ISO 16750-3, section 4.1.2.2.2.2.

Operating mode	Operating mode II.c
Test duration	Use a test duration of 22 h for each plane of the DUT
Test voltage	12V

Requirement

There shall be no damage or cracks.
The performance specifications must be satisfied.

분류 : 진동 내구성 평가

간단 설명 :

ISO 16750-3 규격 _ 차량 내 장착 위치에 따라 가진 조건이 변경될 수 있음.

가진량이 작성된 표와 표에 따른 그래프 정보가 같이 출력이 필요함.

추가 정보

진동 시험 종류

Random Vibration / Sinusoidal Vibration / Shock Vibration